

AMD DEVELOPER HACKATHON · ACT II · UNICORN TRACK

The autonomous underwriter that finds the path to yes.

Crucible puts the loan on trial, writes the conditions, reads the documents, works the phones, and rebuilds denials — every claim cited, on AMD.

Team Milad + Joseph On AMD Instinct MI300X + Fireworks Crucible

THE PROBLEM

Underwriting is a black box that says **no**.

A borrower applies, waits weeks, and a hidden process returns a verdict with no reasoning and no recourse. A denial is a dead end — not an instruction.

OPAQUE

Borrowers and even LOs can't see why a loan was suspended or denied.

COSTLY

Fallout, re-work, and lost-but-approvable loans bleed lenders.

UNACCOUNTABLE

"The model said no" won't survive the coming rules on AI lending.

THE INSIGHT

Make it a **glass box**.

One real loan, argued in the open. Every claim cited — or it isn't made.

➤ ADVERSARY

Prosecutes

Files every reason to deny — each tied to the exact guideline it violates.

ADVOCATE ▼

Defends

For each charge, the strongest guideline-valid path: a product, an investor, an action.

▼ ADJUDICATOR

Rules

Weighs the cited record, abstains when uncited, and issues a ranked Path to Yes.

LIVE · WATCH IT WORK

One real file, **right now.**

PROSECUTION

DEFENSE

PATH TO YES



[Open the real console ↗](#) console requires the running container (scripts/serve.py or docker compose up)

THEN IT WORKS THE FILE

A verdict is where underwriting **starts** — so the agent keeps going.

CONDITIONS

Writes the real report

Master · TRAC · Underwriter II (PTD) · Disclosures — codes, categories, and cleared status, exactly like a live UWM conditions report.

DOCUMENT INTELLIGENCE

Reads & clears — with judgment

Classifies incoming docs, extracts the numbers, and auto-clears conditions with sufficiency checks. A \$900 statement does *not* clear a \$31,400 funds condition.

AUTONOMOUS OUTREACH

Works the phones

Routes every open condition to the right human — title company, insurance agent, broker, borrower — drafts the ask, dispatches via Twilio.

AUS RAILS

Submits the file

Builds the DU-style casefile and submits through the Fannie/Freddie seam — honestly labeled simulated until vendor onboarding.

THE SIGNATURE MOVE

Denied? It **rebuilds the deal.**

When a file can't fund as submitted, the solver searches the allowable moves — product switch, temporary buydown, down payment, debt paydown — and returns the minimal-effort restructure that flips it.

AS SUBMITTED

DTI 55.0% — NOT YET

Alicia Reyes: DTI over the conventional ceiling. Every other system stops here.

REBUILT BY THE AGENT

2-1 buydown → FUNDABLE

Qualifying payment drops, DTI clears, and the change-of-circumstance conditions (0571) are written automatically.

Honest by design: when no realistic restructure exists, it says so — and refers to manual structuring.

THE COMPOUNDING ASSET

Every file makes it **smarter**.

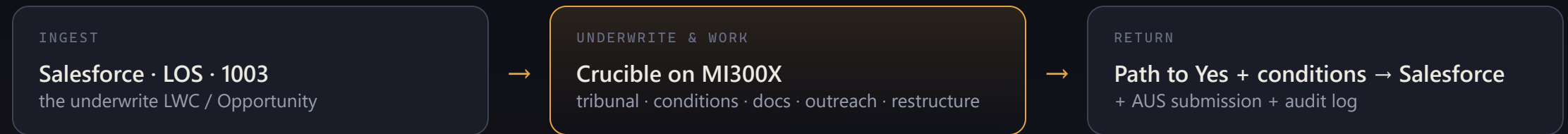
- **Approval scorer** — online weights update on every underwrite; the score sharpens with the book.
- **Restructure memory** — which moves actually flip denials, ranked by real success rate.
- **Clearance map** — which documents clear which conditions, learned from every clearance.

Explainable online learning today — the seam
where a model fine-tuned on your book, trained
on the MI300X, drops in next.

HOW IT WORKS · A BACKEND SERVICE

Your system of record, underwritten on AMD, written back.

Crucible is a headless, containerized service. The web console is the underwriter's window — most traffic is API-to-API.



THE TECHNOLOGY · WHY AMD

A hybrid router is the spine.

High-volume, sensitive, deterministic work stays on-device; only the hard reasoning escalates.

LOCAL LANE · MI300X (ROCm/VLLM)

Guideline RAG · DTI/LTV math · adversary scan · **document parsing** — **borrower PII never leaves the box** · a fine-tuned guideline specialist.

CLOUD LANE · FIREWORKS

Advocate strategy synthesis and final adjudication — only on low-confidence escalation.

~60% of calls handled on the Instinct GPU — private, transparent, **and** cheaper than an all-frontier pipeline.

BUSINESS VALUE · THE MOAT

Denials become recovered revenue — and a defensible record.

REVENUE

Path, not dead end

Convert would-be denials into structured, fundable approvals.

COMPLIANCE

Reg-B by design

Cited adverse-action reasons, generated automatically — cited or the model abstains.

TRUST

Full audit trail

Every decision replays the exact reasoning — what regulators will require of AI underwriting.

ROADMAP · ONE ENGINE, A PLATFORM

The wedge is an autonomous underwriter.

- **Real rails** — DU/LPA vendor onboarding, LOS integrations, voice outreach on the existing dialer stack.
- **Living Loan** — run continuously; ping the borrower the moment a refi or PMI drop becomes fundable.
- **Fiduciary Advocate** — shop investors and products for the borrower, conflicts disclosed.

Same engine. From a wedge to a platform.

CRUCIBLE

*The black box says **no**.
Crucible works the file until it's a **yes**.*

Built on AMD Instinct MI300X + ROCm + Fireworks Team Milad + Joseph Live: tribunal · conditions · doc intelligence · outreach · AUS · learning